apparatus and the system just happens by unitary evolution of the state-vector.

The only problem is that, in a certain sense, we have merely delayed the difficulty. It is not very satisfying to be told that the apparatus "knows" the spin state unless the experimenter—let's say Alice—is allowed to look at the apparatus. Isn't it true that when she does so, she will collapse the wave function of the composite system? Yes and no. For all of Alice's purposes, yes; she will conclude that the apparatus, and the spin, are in one of the two possible configurations and will proceed accordingly.

But now let's bring Bob into the picture. So far, he has not interacted with the spin, the apparatus, or Alice. From his point of view, all three form a single quantum system. No wave function collapse took place when Alice looked at the apparatus. Instead, Bob says that Alice became entangled with the other two component systems.

That's all well and good, but what happens when Bob looks at Alice? For his purposes, he has collapsed the wave function. But then there is good old Charlie …

Does the last entity to look at the system collapse the wave function, or does it just get entangled? Or is there a last looker? I won't try to answer these questions, but what should be apparent is that quantum mechanics is a consistent calculus of probabilities for a certain kind of experiment involving a system and an apparatus. We use it, and it works, but when we try to ask questions about the underlying "reality," we get confused.

7.9 ENTANGLEMENT AND LOCALITY

Does quantum mechanics violate locality? Some people think so. Einstein railed against the "spooky action at a distance" (*spukhafte Fernwirkung*) that he claimed was implied by quantum mechanics. And John Bell became almost a cult figure by proving that quantum mechanics is nonlocal.

On the other hand, most theoretical physicists, particularly those who study quantum field theory, which is riddled with entanglement, would claim the opposite: quantum mechanics done correctly ensures locality.